

MagicBench: Diagnosing Visual Agency Loss and Semantic Dependency in Multimodal LLMs

Anonymous ACL submission

Abstract

Multimodal Large Language Models (MLLMs) typically assume linguistic context invariably enhances visual understanding. We challenge this paradigm in **semantic adversarial scenarios**—specifically magic tricks—where narratives deliberately decouple from physical reality. We introduce **MagicBench**, a diagnostic benchmark of **402** videos evaluating MLLMs against hierarchical linguistic interference, adjudicated via the **Physical Constraint Set (PCS)**, a logic-based protocol assessing adherence to physical laws.

Evaluation uncovers a **Semantic Dependency Paradox**: (1) **Semantic Anchoring**: Entity nouns act as anchors aiding localization, paradoxically boosting performance despite false predicates. (2) **Visual Agency Loss**: In semantic vacuums, multimodal performance collapses **12.4%** ($p < 0.01$) below the vision-only *capability probe*. This gap persists under **symmetric prompting**, revealing an **Functional Perception Suppression** where the fusion layer suppresses autonomous visual search without linguistic triggers. Causal interventions via spatial prompting and signal magnification prove that internal reasoning remains functional, confirming a perceptual **Access Bottleneck**. Our findings suggest MLLMs function as “language-guided passive observers,” advocating for **perceptually-independent** architectures that decouple sensory agency from linguistic dominance.

1 Introduction

The evolution of Multimodal Large Language Models (MLLMs) has ushered in “Early Fusion” architectures, where visual and textual tokens are processed jointly in a shared latent space (Alayrac et al., 2022; Team et al., 2024). Driven by instruction tuning (Liu et al., 2023), the community largely operates under the “*Context is King*” paradigm, assuming that richer linguistic context

monotonically enhances visual grounding (Wang et al., 2023). This creates an inductive bias toward **Linguistic Over-reliance**, presuming a cooperative relationship between modalities while overlooking the risk of **Asymmetric Modality Dominance**, where linguistic priors may prematurely gate or suppress visual evidence.

However, this assumption fractures in adversarial scenarios where language is deceptive or absent. While prior research quantified **hallucination** in static contexts (Yin et al., 2024; Li et al., 2023), dynamic events require models to maintain **Autonomous Physical Entailment (APE)**—the capacity to prioritize sensory evidence and track causal state changes (e.g., object permanence) against a contradictory narrative. Current benchmarks (Yu et al., 2023; Maaz et al., 2024) often lack this causal complexity, failing to disentangle whether a model is truly “seeing” the physics or merely defaulting to linguistic shortcuts (Leng et al., 2024; Golovanevsky et al., 2025).

To probe these boundaries, we propose **Magic** as a uniquely rigorous testbed for APE. A magician’s “patter” is designed to decouple symbolic narratives from physical reality, exploiting cognitive blind spots to mislead the observer. Magic is not a mere novelty; it operationalizes **causal state tracking** and **cross-modal conflict resolution** in their purest adversarial forms. We introduce **MagicBench**, a diagnostic benchmark of **402** high-fidelity videos categorized by hierarchical linguistic interference (Lies, Misdirection, Silence). Adjudicated via a **Physical Constraint Set (PCS)** grounded in First-Order Logic, our evaluation uncovers a counter-intuitive **Semantic Dependency Paradox**.

This paradox manifests in two distinct effects. First, **Semantic Anchoring** (the *Spotlight Effect*) occurs in deceptive scenarios, where entity nouns (e.g., “coin”) act as anchors that facilitate localization; the gain from object grounding outweighs the interference from false predicates (lies). Sec-

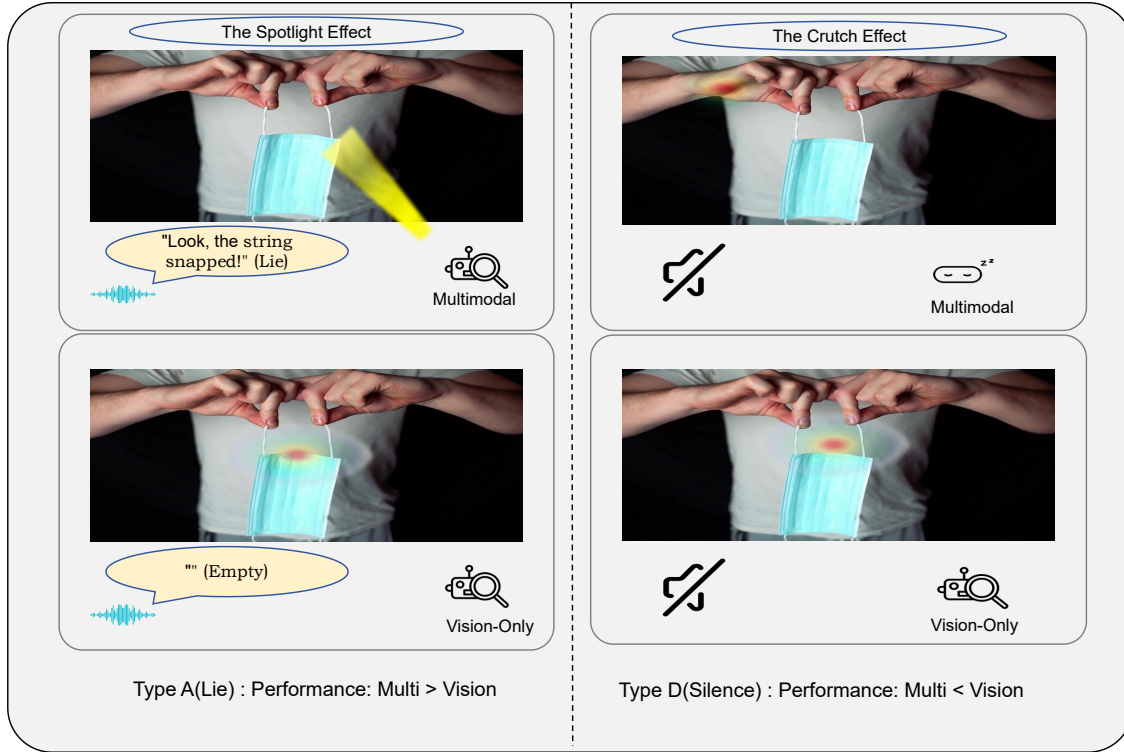


Figure 1: **The Semantic Dependency Paradox. (Left) The Spotlight Effect (Semantic Anchoring):** In deceptive scenarios, entity nouns (e.g., “string”) act as a *semantic anchor*. Although the predicate is false (“snapped”), the anchor provides a top-down spatial prior that facilitates object localization, leading to successful trick demystification. **(Right) The Crutch Effect (Visual Agency Loss):** In semantic vacuums (Silence), the multimodal model suffers from a catastrophic collapse in performance. Lacking a linguistic trigger to orient attention, the model’s visual encoder enters a state of **perceptual idling**. In contrast, the *Vision-Only* capability probe successfully attends to the finger-loop maneuver, revealing a systemic dependency on linguistic scaffolding in current early-fusion MLLMs.

ond, and more alarmingly, we identify a **Visual Agency Loss** (the *Crutch Effect*) in semantic vacuums (Silence). Here, multimodal performance collapses **12.4%** ($p < 0.01$) below the *Vision-Only* capability probe. This gap persists under **Symmetric Intensity Ablation (SIA)**, where prompts are identical, revealing that current MLLMs function as “language-guided passive observers” that enter a state of **Perceptual Idling** without linguistic triggers.

Beyond observation, we perform **Interventional Diagnosis**. Visual prompting (“Red Circle”) and signal enhancement tests confirm that failures stem from an **Access Bottleneck** rather than reasoning deficits. This aligns with mechanistic evidence of **functional suppression** of visual tokens within transformer blocks (Liu et al., 2025; Nikankin et al., 2025). Our contributions include: (1) **MagicBench**, a 402-video diagnostic benchmark for semantic adversarial robustness; (2) the discovery of the **Spotlight & Crutch** duality; (3) the causal identification of a systematic **Access Bottleneck** in early-

fusion MLLMs; and (4) empirical evidence for **Perceptually-Independent** architectures to decouple sensory agency from linguistic dominance.

2 Related Work

2.1 Physical Entailment and Prompt-Induced Hallucination

Hallucination research has transitioned from static object existence (Li et al., 2023) to temporal consistency in video domains (Maaz et al., 2024; Li et al., 2025). However, most benchmarks focus on summarization-level events. MagicBench introduces a more granular challenge: **sub-second state tracking** under adversarial narratives. We build upon the framework of **Prompt-Induced Hallucination** (Rudman et al., 2026), where deceptive linguistic context effectively “blinds” the model to physical evidence. Unlike synthetic benchmarks like CLEVRER (Yi et al., 2019), we test **Physical Entailment** in real-world scenarios, requiring models to maintain causal consistency against narratives that actively contradict visual kinematics.

2.2 Modality Conflict and Competitive Circuits

The competition between pixels and priors is a focal point of recent mechanistic studies. While **Visual Override** (seeing overrides knowing) is possible in simple tasks (Ortu et al., 2025; Golovanevsky et al., 2025), we demonstrate that this mechanism is highly fragile in dynamic, adversarial contexts. This aligns with recent circuit-level evidence: Nikankin et al. (2025) and Hua et al. (2025) reveal that MLLMs employ distinct, competing circuits for different modalities, often leading to the **active suppression** of visual signals when linguistic priors are strong. Our discovery of *Visual Agency Loss* provides macro-behavioral evidence for the “Bad Eyes” phenomenon (Aravindan et al., 2025), suggesting that critical visual information is functionally lost or misrouted during cross-modal fusion within Transformer blocks (Liu et al., 2025).

2.3 Active Perception and Visual Steering

Standard benchmarks often allow models to succeed via passive recognition. In contrast, the theory of **Active Perception** (Bajcsy et al., 2018) posits that vision is an exploratory activity. MagicBench operationalizes this by identifying a form of “perceptual idling” in current architectures. To mitigate such deficits, recent works explore **Visual Task Prompting** (VTPrompt) (Hong¹ et al., 2024) and **Image-of-Thought** (IoT) pipelines (An et al., 2024) to steer attention via spatial markers or sequential auditing. Systematic frameworks like **VP-Bench** (Xu et al., 2025) evaluate these steering capabilities in cooperative settings; MagicBench extends this to **Semantic Adversarial** contexts, demonstrating that while attention-steering (Shi et al., 2023) provides marginal gains, the underlying modality dependency remains an unresolved architectural bottleneck.

3 MagicBench Construction

3.1 Data Curation and APE Capabilities

We curate **MagicBench**, a diagnostic benchmark comprising **402** high-fidelity magic videos (>3.2 hours). We define the core capability operationalized by MagicBench as **Autonomous Physical Entailment** (APE)—the capacity to maintain a consistent causal world model when linguistic narratives actively contradict visual kinematics. Magic serves as a uniquely rigorous testbed for APE because the

“patter” is designed to decouple symbolic narratives from physical reality.

Taxonomy and Annotation Rigor. The dataset is categorized by hierarchical linguistic interference: (1) **Type A (Direct Lie, $N = 146$)**: Explicit falsification of the visual state. (2) **Type B (Misdirection, $N = 91$)**: Attentional diversion to non-critical regions. (3) **Type C (Patter, $N = 83$)**: High-entropy irrelevant conversational filler. (4) **Type D (Silence, $N = 82$)**: A critical control group representing a **Semantic Vacuum**.

Each video is grounded in a **Physical Constraint Set (PCS)**—a formalization of objective facts revealed in tutorial segments. To ensure logical fidelity, three experts independently drafted physical steps, resolved via a **Consensus Protocol** to maximize the recall of sub-second maneuvers.

To quantify objectivity, a reliability audit on 30 samples yielded a **BERTScore F1 of 0.87** (Zhang et al., 2019), confirming that experts captured the same underlying logic despite phrasing variations. On average, each PCS instance comprises 5.7 ± 1.4 **atomic physical constraints**, with a mean length of 64.3 ± 8.5 **tokens**. MagicBench is committed to Open Science; the full dataset and symbolic facts will be publicly released.

3.2 Sampling and Sensory Sufficiency

Magic tricks hinge on transient maneuvers. We employ an **Adaptive Hierarchical Sampling** strategy (4fps for videos < 20s) to capture these cues within MLLM context limits.

Recall and Existence Proof. A human audit on 60 videos confirms that critical manipulation frames were captured in **98.3%** of cases (95% CI: [91.1%, 99.7%]). To rule out temporal resolution as a confound, we utilize the **Vision-Only Capability Probe** as an **internal control**. Since both settings use identical 4fps inputs, the robust performance of the Vision-Only probe (e.g., 50.7% CPA for Gemini-2.5-Pro) serves as an **existence proof** that the sampled frames contain sufficient information. Any performance collapse in multimodal settings is thus a failure of *cognitive integration*, not *sensory acquisition*.

3.3 Evaluation Protocol: Physical Entailment

Standard benchmarks rely on textual similarity, which fails to capture causal logic. We propose

Step 1: The Zero-Tolerance Veto (Pre-requisite)			
Score $\rightarrow 0$ if: (1) Prop Hallucination; (2) State Contradiction; (3) Magical Thinking.			
Step 2: Scoring Rubric (Scale 0 – 10)			
Metric	High (7–10)	Mid (4–6)	Low (0–3)
VTG	Precise. Correctly identifies Body Part, Specific Action, and Temporal Ordering aligned with PCS.	Partial. Identifies Body Part but is vague on Action or Timing; or correct moment but wrong location.	Fail. Wrong location, hallucinates visual events, or provides completely vague descriptions.
CPA	Entailed. Explanation is physically entailed by PCS (Terminology Agnostic). Describes correct mechanics.	Plausible. Identifies general category (e.g., Sleight of Hand) but misses key mechanical steps.	Fail. Triggered Hard Failure Veto OR explains a completely wrong mechanism.
CFR	Resilient. Explicitly refutes the audio lie OR provides a physical explanation contradicting the lie.	Skeptical. Expresses doubt (e.g., "unlikely") but fails to provide a definite physical alternative.	Fooled. Accepts the illusion or lie as fact (e.g., "it melted") or invents props to justify the lie.

Table 1: **The PCS-based Judgment Protocol.** Models must first pass the *Zero-Tolerance Veto*. If passed, they are graded on a 0–10 scale across three dimensions based on the degree of alignment with the Physical Constraint Set (PCS).

an **LLM-as-a-Judge** protocol grounded in **Physical Entailment**, assessing reasoning against three FOL invariants: **Object Permanence**, **Kinematic Continuity**, and the **Visual Supremacy Axiom** (Appendix C).

Blind-Logic Adjudication. To prevent circular reasoning, our protocol is **strictly decoupled**: the judge receives only the model’s textual output and symbolic PCS facts, but never the raw video. This transforms the judge from a subjective observer into a deterministic logic checker. Grading follows a two-stage rubric: a **Zero-Tolerance Veto** for physical impossibilities, followed by a **Scalar Rubric** (0–10) for completeness (see Table 1).

Reliability and Bias Mitigation. To ensure statistical rigor and address *self-preference bias*, we expanded human validation to a stratified sample of $N = 80$ (20% of the dataset). As shown in Table 6, our protocol exhibits exceptional reliability: (1) **Cross-Architecture Objectivity:** An independent judge (*Claude-3.5-Sonnet*) achieved high correlation with human experts (Avg. $r = 0.8398$), mirroring GPT-4o’s performance ($r = 0.878, p < 0.001$). This confirms adjudication is driven by **symbolic PCS axioms** rather than model identity. (2) **Targeted Veto Audit:** We evaluated 20 cases where model explanations were fundamentally unphysical (vetted as CPA=0 by Ground Truth). Without PCS grounding, a **vanilla GPT-4o judge** exhibited an **85.0% False Positive Rate (FPR)**, accepting physical impossibilities as plausible (Table 2). Our PCS-grounded judge maintained a **0.0% FPR**. (3) **Noise and Persona Robustness:** On $N = 50$ sam-

Judge Setting	Avg. Score ($\uparrow$)	FPR ($\downarrow$)
Vanilla GPT-4o (No PCS)	5.3/10	85.0%
PCS-Grounded (Ours)	0.0/10	0.0%

Table 2: Ablation of PCS grounding ($N = 20$). FPR denotes the rate of non-zero scores assigned to physically impossible explanations. The PCS protocol successfully mitigates the judge’s inherent bias toward accepting illusory narratives.

ples, introducing 10% spatial jitter into PCS coordinates yielded negligible variance (**MAPE=5.4%**, $p = 0.38$). Across the **entire dataset** ($N = 402$), scores from distinct judge personas achieved a **Pearson $r = 0.7949$** ($p < 0.001$), with 100% identical veto decisions.

3.4 Metrics

We report performance across three dimensions: (1) **Visual-Temporal Grounding (VTG):** Spatial and temporal localization precision. (2) **Causal-Physical Accuracy (CPA):** Adherence to objective physical laws (the primary APE metric). (3) **Counterfactual Resilience (CFR):** Robustness against the compounded effect of visual trickery and linguistic deception.

4 Experiments

4.1 Experimental Setup

Models. We evaluate three state-of-the-art MLLMs representing diverse architectures: (1) **GPT-4o:** the current commercial SOTA for reasoning; (2) **Gemini-2.5-Pro:** a native multimodal model; and (3) **Qwen2.5-VL-72B:** the leading

Model	Setting	Type A (Lie)	Type B (Misdir.)	Type C (Patter)	Type D (Silence)	Avg.
GPT-4o	Vision-Only (Probe)	33.0 $\pm$ 1.1	38.1 $\pm$ 1.5	34.5 $\pm$ 1.2	28.2$\pm$1.0	33.7
	Multi-Forensic (SIA)	43.4$\pm$1.3	43.3$\pm$1.6	33.6 $\pm$ 1.1	15.8 $\pm$ 0.9 $\uparrow$	38.1
Gemini-2.5-Pro	Vision-Only (Probe)	47.3 $\pm$ 1.4	46.3 $\pm$ 1.2	46.3 $\pm$ 1.5	50.7$\pm$1.3	47.6
	Multi-Forensic (SIA)	42.1 $\pm$ 1.4	45.8 $\pm$ 1.2	38.4 $\pm$ 1.1	37.8 $\pm$ 1.1 $\uparrow$	41.0
Qwen2.5-VL	Vision-Only (Probe)	32.4 $\pm$ 1.0	34.5 $\pm$ 1.3	35.7$\pm$1.2	30.6$\pm$1.1	33.3
	Multi-Forensic (SIA)	48.7$\pm$1.4	40.3$\pm$1.6	32.3 $\pm$ 1.0	22.7 $\pm$ 1.2 $\uparrow$	36.0
<i>Upper Bound(GPT-4o)</i>	Truth Context	86.7	87.0	87.0	86.3	86.8

Table 3: **Visualization of the Semantic Dependency Paradox across three SOTA MLLMs.** (a) In deceptive scenarios (Type B), linguistic entities act as semantic anchors that facilitate visual grounding, leading to the *Spotlight Effect*. (b) In semantic vacuums (Type D), the absence of linguistic triggers causes a catastrophic collapse in performance relative to the models’ intrinsic visual capacity, revealing a chronic *Visual Agency Loss* (the *Crutch Effect*). Error bars represent the Standard Error (SE) estimated via 1,000 bootstrap iterations.



Figure 2: **Visualization of the Semantic Dependency Paradox.** (a) **The Spotlight Effect (Type B):** Linguistic entities act as semantic anchors that facilitate visual grounding, enabling multimodal models to surpass vision-only baselines. (b) **The Crutch Effect (Type D):** In semantic vacuums, all models exhibit a catastrophic collapse relative to their intrinsic capacity, revealing a chronic *Visual Agency Loss*. Red arrows highlight the irrecoverable performance gap induced by the multimodal fusion layer.

open-source MLLM. All inferences are conducted at a temperature of 0.6.

Input Specification. To isolate semantic reasoning from low-level auditory perception errors, we utilize **ground-truth textual transcripts** as the proxy for the audio modality. In Type D (Silence), the audio field is an **empty string**, while the system prompt continues to mandate multimodal cross-referencing.

Symmetric Intensity Ablation (SIA). To ensure a fair modality-level comparison and address the confound of instruction design, we design five settings:

- **Vision-Only (Capability Probe):** The model receives only video frames with a high-intensity forensic prompt, instructing it to ignore implied audio and rely strictly on pixels.
- **Multimodal (Neutral):** Default behavior us-

ing generic descriptive instructions.

- **Multimodal (Skeptic):** Content-focused intervention explicitly warning the model of deception.
- **Multimodal (Forensic) [SIA]:** Our primary experimental group, which utilizes a prompt **functionally symmetric** to the Vision-Only probe. We quantify this symmetry by extracting the set of core logical and stylistic constraints from both prompts, achieving an **Instruction Overlap F1 score of 0.9333**. This Symmetric Intensity Ablation (SIA) protocol holds instruction intensity constant, effectively isolating the modality channel as the sole independent variable (see Appendix Table 8 for the side-by-side comparison).
- **Truth Context:** Deceptive transcripts are replaced with ground-truth physical principles

to establish the theoretical reasoning ceiling.

4.2 Main Results: The Semantic Dependency Paradox

Table 3 summarizes the performance across all models and types. We report three primary observations based on Causal-Physical Accuracy (CPA):

1. Semantic Anchoring (The Spotlight Effect).

In deceptive scenarios (Type A/B), the **Multimodal (Forensic)** setting consistently surpasses the *Vision-Only* baseline across all models. For instance, GPT-4o improves from 33.0% to 43.4% in Type A, and Qwen2.5-VL surges from 32.4% to 48.7%. As proven in our component ablation (Sec 5.1), this gain is driven by **entity nouns acting as semantic anchors** that narrow the visual search space, allowing models to localize objects despite false predicates.

2. Performance Inversion in Silence (The Crutch Effect).

In Type D (Silence), we observe a catastrophic collapse in multimodal performance. Despite receiving identical forensic instructions (SIA), all models exhibit a **statistically significant Residual Gap** compared to the unimodal probe (e.g., **12.4% for GPT-4o**, $p < 0.01$). This confirms that the mere presence of an inactive audio channel induces **Visual Agency Loss**: the visual encoder defaults to a state of *Perceptual Idling* when its linguistic “crutch” is removed.

3. Cross-Model Stability of the Paradox. As visualized in Fig. 2, the paradox is an architectural pathology common to early-fusion MLLMs. While the magnitude of the Spotlight gain varies by model—reflecting different modality-weighting strategies—the **Crutch Effect is universally stable**. Even the high-performing Gemini-2.5-Pro suffers a 12.9% drop in Type D relative to its own intrinsic capacity.

4.3 Causal Evidence: Spatial Specificity

To quantify the *access bottleneck* hypothesis, we performed the spatial intervention (Fig. 3) on $N = 60$ failure cases. By highlighting ground-truth coordinates defined by the PCS, reasoning accuracy (CPA) for Qwen2.5-VL recovered from **16.8% to 29.8%**. Mechanistically, this recovery coincides with a surge in **Relative Spatial Density (RSD)** from **0.47 (avoidance) to 2.66 (active grounding)** within the critical region. This proves that failures stem from **Perceptual Access Gating** rather than a lack of reasoning competence.

5 Diagnosis: The “Spotlight & Crutch” Framework

We analyze the behavioral mechanisms driving the *Semantic Dependency Paradox*. Our results establish that current MLLMs operate under a **Linguistic-Driven Saliency Hierarchy**, where visual reasoning is strictly contingent on linguistic activation.

5.1 Dissecting the Spotlight: Anchoring vs. Alerting

In deceptive scenarios, multimodal performance paradoxically improves over the unimodal baseline. For instance, in Type B (Misdirection), the *Multimodal (Skeptic)* setting achieves **44.0%**, surpassing the *Vision-Only* probe of **38.1%**. We attribute this to **Semantic Anchoring**: while linguistic *predicates* are false (e.g., “melted”), the *entities* (e.g., “coin”) act as spatial priors that narrow the visual search space.

Mechanism Isolation: Component Ablation.

To verify whether this gain stems from specific semantic guidance or generic attentional arousal, we conducted a controlled ablation on $N = 50$ Type A samples (Table 4). The results reveal a cognitive hierarchy: (1) **Alerting**: Generic cues (“Look here”) restore performance only to the visual baseline ($33.6\% \approx 33.0\%$), proving that simple audio presence overcomes silence-induced inertia but cannot guide the visual encoder. (2) **Anchoring**: The significant boost is exclusively driven by specific nouns (44.2%). The model utilizes these anchors for localization while filtering out false predicates, confirming an **entity-centric alignment bias**. Conversely, high-entropy noise (Type C) degrades performance due to **Attention Dispersion**.

Text Condition	Function	CPA (%)	vs. Baseline
Vision-Only (Baseline)	—	33.0	—
Generic Cues (“Look here”)	<i>Alerting</i>	33.6	+0.6 (n.s.)
Type C (Irrelevant Patter)	<i>Interference</i>	25.6	↓ 7.4
Entity Keywords (“Coin”)	<i>Anchoring</i>	44.2	↑ 11.2
Full Deceptive (Original)	<i>Conflict</i>	43.4	↑ 10.4

Table 4: Component ablation in Type A scenarios ($N = 50$). The null result for Generic Cues confirms that the Spotlight Effect is driven by noun-based anchoring.

5.2 Visual Agency Loss (The Crutch Effect)

The collapse in Type D (Silence) provides definitive evidence for **Visual Agency Loss** (i.e., modality-

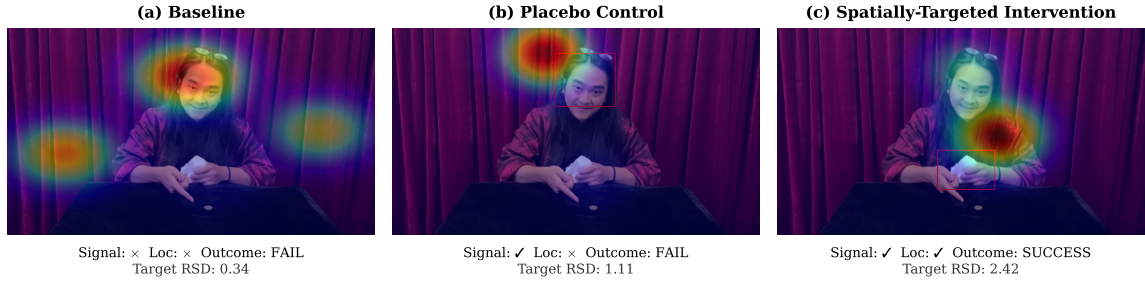


Figure 3: **Causal Diagnosis of the Access Bottleneck via Spatial Intervention.** We isolate the mechanism of Visual Agency Loss using a triplet control ($N = 60$ for Qwen2.5-VL). **(a) Baseline (Hijacked):** Deceptive audio diverts attention from critical regions (mean RSD ≈ 0.47), causing reasoning failure. **(b) Placebo Control:** Highlighting non-critical regions fails to restore CPA, ruling out generic attention-arousal artifacts. **(c) Targeted Intervention:** Precise spatial prompting (analogous to oracle *VTPrompt* (?)) restores CPA (16.8% $\rightarrow$ 29.8%) and surges RSD (to 2.66). These results localize the failure to a perceptual *access bottleneck* rather than a reasoning deficit. Detailed analysis provided in Sec. 5.3.

driven perceptual inertia). By contrasting *Capacity* (Vision-Only) and *Behavior* (Multimodal), we isolate a profound gap: the robust score of **28.2%** establishes that models *possess* the intrinsic hardware to decode the trick, yet this capability remains **dormant** (4.4%) under standard interaction. This confirms the **Language-Guided Observer Hypothesis**: due to instruction-tuning biases, models default to a state of **Perceptual Idling** without linguistic triggers, providing macro-behavioral evidence for the “**Bad Eyes**” phenomenon (Aravindan et al., 2025) where visual information is functionally misrouted during cross-modal fusion (Liu et al., 2025; Nikankin et al., 2025).

5.3 Causal Diagnosis: Restoring Perceptual Access

To disentangle whether failures stem from reasoning deficits or an access bottleneck, we conduct two complementary interventions: **Exogenous Spatial Prompting** (top-down) and **Signal Enhancement** (bottom-up).

Restoring Access. Highlighting critical coordinates (Red Circle) for $N = 60$ failure cases caused Qwen2.5-VL’s CPA to recover from **16.8% to 29.8%**, coinciding with a surge in **Relative Spatial Density (RSD)** from **0.47 to 2.66**. Furthermore, we performed a **Targeted Stress Test** on $N = 20$ “High-Discrepancy” samples (Multimodal CPA < 3.0 vs. Vision-Only CPA > 7.0). By magnifying critical regions, the CPA for these samples surged from **11.26% to 63.2%** ($p < 0.001$). This yields a critical insight: **Visual Override** (Ortu et al., 2025) is possible but requires a disproportionately high perceptual threshold in adversarial

contexts. In standard settings, the fusion layer imposes an **Functional Perception Suppression** that suppresses visual features.

5.4 Disentangling Confounding Factors

We address concerns regarding token budgets and prompt artifacts through two logical controls: (1) **Resource Control:** In Type D, the audio channel is an empty string, eliminating token congestion. The fact that performance is *lowest* here rules out compute budget as the failure driver. (2) **Symmetric Intensity Ablation (SIA):** Comparing the *Vision-Only* probe to the *Multimodal (Forensic)* setting using **identical high-intensity prompts** reveals a persistent **12.4% Residual Gap**. This proves that Visual Agency Loss is an **architectural pathology** rather than a transient prompting artifact.

5.5 Limits of Inference-Time Mitigation (IoT)

Finally, we investigate whether advanced inference strategies can resolve this pathology. We implement an **Image-of-Thought (IoT)** pipeline (An et al., 2024), mandating a visual inventory before reasoning.

Strategy	Type A	Type B	Type C	Type D	Avg.
Standard (Neutral)	35.6	34.5	25.6	4.4	25.0
Forensic (SIA)	43.4	43.3	33.6	15.8	34.0
IoT (Inference)	46.2	45.8	36.4	21.2 $\uparrow$	37.9
Vision-Only	33.0	38.1	34.5	28.2	33.5

Table 5: **Inference Strategy Comparison** ($N = 402$). While IoT improves performance, it fails to close the 7.0% gap with the Vision-Only probe in Type D ($p < 0.01$).

As shown in Table 5, while IoT achieves peak

multimodal performance (**46.2%** in Type A) by synergizing with semantic anchors, it **fails to resolve the gap** in semantic vacuums (**21.2% vs. 28.2%**). This suggests that modality dependency is rooted in the **functional suppression** of visual tokens at the integration layer, a tax that cannot be fully bypassed by inference-time heuristics.

6 Discussion: Toward Perceptual Independence

The *Semantic Dependency Paradox* challenges the “Context is King” dogma. Our findings prove that current MLLMs are not integrated reasoners but **language-guided passive observers** that prioritize linguistic priors over empirical sensory evidence.

6.1 Architectural Pathology: The Fusion Bottleneck

Our behavioral diagnosis acts as a macro-scale validation of the **competing circuits** identified by Nikankin et al. (2025) and the **functional suppression** described by Hua et al. (2025). The failure of even the advanced IoT pipeline (Sec 5.5) to bridge the gap to the Vision-Only probe localizes the bottleneck to the **representation-fusion level**. This confirms a Functional Pathology in early-fusion architectures: high-entropy narratives effectively “blind” the visual encoder. We propose a transition to **Decoupled Visual-First Inference**, establishing a **Pure Perception Stage** to extract objective physical factoids in a semantic vacuum before linguistic filtering.

6.2 Adversarial Alignment and Epistemic Courage

The *Crutch Effect* identifies an emergent pathology of instruction-tuning: **Modality Bias** (Nikankin et al., 2025). By training on perfectly aligned alt-text data, MLLMs learn to treat language as a reliable shortcut, resulting in a failure of *Active Perception* when linguistic guidance is absent. To cultivate independent agency, the training paradigm must move toward **Adversarial Modality Alignment**.

We advocate for incorporating a significant ratio of **mismatched semantic pairs** and silent videos during pre-training. This would force models to resolve cognitive dissonance by prioritizing visual verification, thereby “hardening” the perceptual pipeline. A robust agent must possess the **Epistemic Courage** to falsify a linguistic narrative

when it contradicts the immutable physical laws of pixels.

6.3 Beyond Magic: A Canary Task for Safety

While magic is domain-specific, the *Visual Agency Loss* it exposes is a universal predictor of failure in high-stakes environments. MagicBench serves as a “**Canary Task**” for safety-critical deployment: if a model cannot maintain *Autonomous Physical Entailment* (APE) against a magician’s patter, it is structurally vulnerable to **Cognitive Hijacking** in autonomous driving (e.g., hacked navigation commands) or disinformation defense. We argue that **Perceptual Autonomy** must become a core metric in future safety benchmarks.

7 Conclusion

In this work, we introduced **MagicBench**, a diagnostic framework designed to probe the cognitive boundaries of Multimodal LLMs through the lens of semantic adversarial conflicts. Our systematic evaluation uncovers a fundamental **Semantic Dependency Paradox**: while linguistic anchors can facilitate object localization (the *Spotlight Effect*), the absence of these anchors triggers a catastrophic collapse in performance (the *Crutch Effect*), revealing a chronic **Visual Agency Loss**.

By utilizing our *Vision-Only* setting as a **Capability Probe**, we quantified a staggering gap between latent perceptual capacity and operational multimodal behavior. Our interventional analyses—ranging from spatial prompting to signal enhancement—prove that this failure is not a reasoning deficit, but a **Perceptual Access Bottleneck** where visual features are functionally suppressed by linguistic dominance. Crucially, we demonstrate that even advanced inference-time strategies like *Image-of-Thought* cannot bridge this gap, localizing the pathology to the early-fusion integration layer.

The implications extend far beyond magic. We propose MagicBench as a vital “**Canary Task**” to measure a model’s **Perceptual Autonomy** before deployment in safety-critical domains. To mitigate Cognitive Hijacking, future architectures must shift from passive alignment toward **Perceptual Independence**, treating linguistic input as a fallible hypothesis to be adjudicated against the immutable ground truth of pixels.

Limitations

While **MagicBench** identifies a systematic pathology in MLLM perception, we acknowledge several constraints that define the scope of our findings:

Ecological Validity and Domain Generalization. Our study utilizes magic as a proxy for **Semantic Adversarial Attacks**. While magic represents a “worst-case” stress test for cognitive conflict, the visual distribution (e.g., stage lighting, specific props) differs from safety-critical domains like autonomous driving. Future work is required to verify if the *Semantic Dependency Paradox* manifests with equal severity in scenarios with lower semantic density but higher safety stakes. We propose that MagicBench serves as a **Canary Task** to measure *Autonomous Physical Entailment* (APE), but it does not replace domain-specific safety evaluations.

Temporal Resolution vs. Context Limits. Despite our **98.3% Sampling Recall** and the *Vision-Only* existence proof (Sec 3.2), the 4fps sampling rate remains an inherent bottleneck. While sufficient for capturing the discrete states required for our PCS logic, extremely rapid micro-sleights or simultaneous multi-object tracking over extended durations may be compromised. This reflects a fundamental trade-off between temporal resolution and the context window limits of current MLLMs. As long-context architectures mature, evaluating MagicBench at native frame rates (e.g., 30fps) will be a critical next step.

Behavioral Proxies vs. Mechanistic Proofs. In our interventional diagnosis (Sec 5.3), we utilized Grad-CAM and RSD maps as behavioral proxies to visualize attention shifts. We caution that these visualizations are aggregative and do not mathematically isolate the specific attention heads or sparse autoencoder features responsible for token suppression. For closed-source models like GPT-4o, our findings remain a clinical behavioral diagnosis. Future research utilizing white-box probing on open-source weights (Nikankin et al., 2025) is needed to map the precise neural circuits that govern the *Crutch Effect*.

The Confound of Alignment Bias. The observed **Visual Agency Loss** may be partially exacerbated by current Supervised Fine-Tuning (SFT) paradigms, which heavily penalize hallucinations by enforcing strict adherence to linguistic prompts.

Our study focuses on deployed models and does not decouple “nature” (base model perceptual capacity) from “nurture” (instruction-following bias). Determining whether the *Crutch Effect* is an architectural inevitability of early-fusion or an artifact of the “Context is King” training objective remains an open scientific question.

Adjudication Ceiling. While our PCS-grounded judge demonstrates high reliability ($r = 0.88$) and robustness to noise, the evaluation protocol still relies on an LLM backbone. Although we mitigate self-preference bias through symbolic facts and cross-model validation (Claude-3.5), the adjudication remains subject to the inherent reasoning ceiling of the judge model itself.

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A Evaluation Protocol Validation

To ensure the reliability of our GPT-4o-based judge and mitigate potential self-preference bias, we conducted a rigorous meta-evaluation on a stratified subset of 80 samples, covering all linguistic types and performance ranges.

Methodology. To establish a robust human baseline, **three expert annotators** independently graded the 80 samples based on the **Physical Constraint Set (PCS)**. The final human score for each sample was derived by averaging the three annotators’ scores to mitigate individual subjectivity. We then compared these consensus human scores with those generated by **GPT-4o (Our Judge)** and **Claude-3.5-Sonnet** (an independent SOTA model)

Results. As shown in Table 6, our GPT-4o judge demonstrates an exceptionally high correlation with human judgment across all three metrics (Pearson $r > 0.86$, $p < 0.001$). Notably, in the critical **CPA** metric, GPT-4o achieves a correlation of **0.87**, surpassing Claude-3.5 ($r = 0.78$). This confirms that GPT-4o aligns more closely with the strict physical constraints defined in PCS. The strong alignment across different evaluators validates that our scoring criteria are objective and reproducible, effectively ruling out stochasticity or model-specific bias.

Metric	Pearson Correlation (r)		
	Human vs. GPT-4o	Human vs. Claude	GPT-4o vs. Claude
VTG (Visual Grounding)	0.8722	0.8669	0.7936
CPA (Physical Accuracy)	0.8825	0.7899	0.7328
CFR (Resilience)	0.8793	0.8627	0.8198
<i>Average</i>	<i>0.878</i>	<i>0.8398</i>	<i>0.7854</i>

Table 6: **Meta-evaluation of the Judge.** The high correlation between Human and GPT-4o (Avg. $r = 0.87$) validates the reliability of our automated evaluation protocol. All correlations are statistically significant ($p < 0.001$).

B Data Quality & Sampling Audit

A critical concern regarding video LLMs is whether sparse sampling strategies capture the fleeting moments essential for magic tricks (e.g., the precise frame of a false transfer). To address this, we conducted a rigorous audit of our **Adaptive Hierarchical Sampling** strategy.

Methodology. We randomly sampled $N = 60$ videos from the MagicBench dataset. For each video, an expert annotator identified the *Critical Time Window* (CTW)—the specific duration where the core manipulation occurs (typically $0.5s \sim 2.0s$). We then verified whether the sampled frames (fed to the model) included at least one clear frame within this CTW.

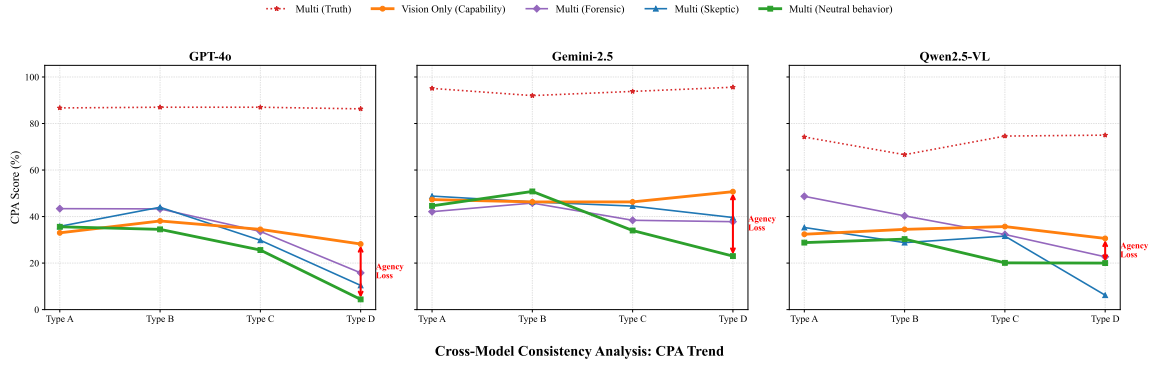
Results. The audit reveals an exceptionally high **Critical Frame Recall Rate** of **98.3%**, with the critical moment successfully captured in 59 out of 60 videos.

$$\text{Recall} = \frac{\text{Captured Samples}}{\text{Total Samples}} = \frac{59}{60} \approx 98.3\% \quad (\text{B.1})$$

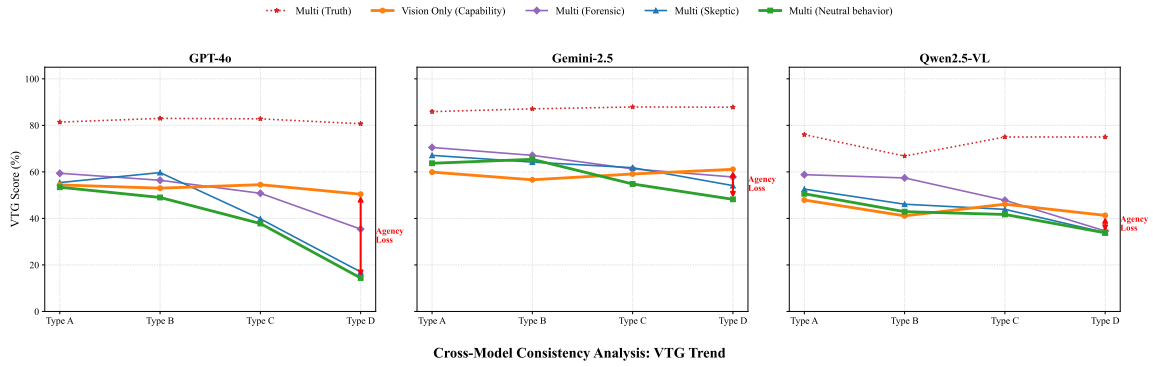
The 95% Confidence Interval (Wilson Score Interval) is [91.1%, 99.7%]. This confirms that in the vast majority of cases, reasoning failures are attributable to the model’s cognitive deficits (e.g., Attention Hijacking or Agency Loss) rather than missing visual evidence caused by downsampling.

C Detailed Experimental Results

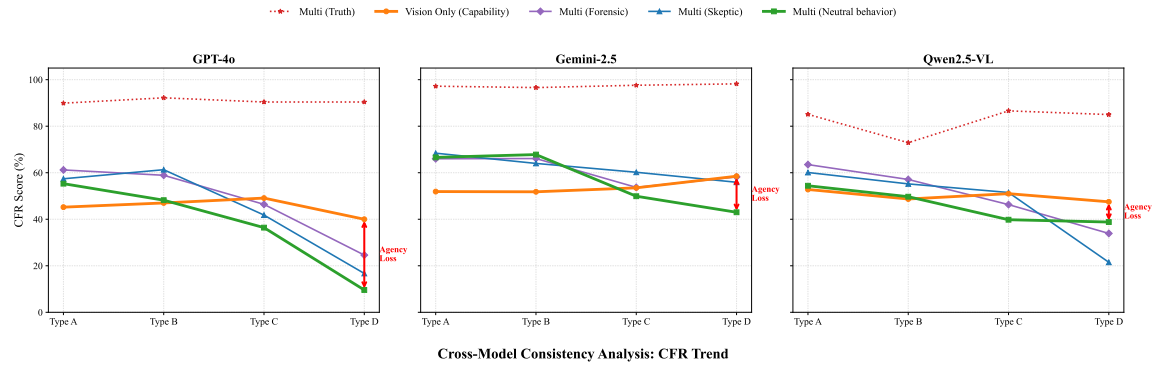
Due to space constraints, the main text reports aggregated performance. Table 7 provides a comprehensive breakdown of Visual-Temporal Grounding (VTG), Causal-Physical Accuracy (CPA), and Counterfactual Resilience (CFR) across all models and linguistic types.



(a) **Causal-Physical Accuracy (CPA) Trends.** Note the consistent "Crutch Effect" in Type D (Silence), where Vision-Only (Orange) outperforms Multimodal (Green/Blue) across all architectures.



(b) **Visual-Temporal Grounding (VTG) Trends.** Visual localization capability also degrades in silence for multimodal settings, indicating that the agency loss is perceptual, not just logical.



(c) **Counterfactual Resilience (CFR) Trends.** Vision-Only models show consistently higher resilience to confusion in the absence of audio cues.

Figure 4: **Cross-Model Consistency Analysis.** We visualize the robustness trends across GPT-4o, Gemini-2.5-pro, and Qwen2.5-VL. The red arrows highlight the **Visual Agency Loss** gap in Type D scenarios, confirming that this is a fundamental architectural pathology common to SOTA MLLMs.

Model	Setting	VTG				CPA				CFR			
		A	B	C	D	A	B	C	D	A	B	C	D
GPT-4o	Vision-Only	54.4	53.0	54.5	50.4	33.0	38.1	34.5	28.2	45.2	47.0	49.1	40.0
	Multi (Neutral)	53.4	49.0	37.8	14.4	35.6	34.5	25.6	4.4	55.3	48.2	36.4	9.6
	Multi (Skeptic)	55.4	59.7	39.8	17.0	35.8	44.0	29.8	10.4	57.4	61.3	41.8	16.7
	Multi (Forensic)	59.4	56.4	50.8	35.4	43.4	43.3	33.6	15.8	61.2	58.9	46.4	24.6
	<i>Truth Context</i>	<i>81.4</i>	<i>83.0</i>	<i>82.8</i>	<i>80.7</i>	<i>86.7</i>	<i>87.0</i>	<i>87.0</i>	<i>86.3</i>	<i>89.9</i>	<i>92.2</i>	<i>90.4</i>	<i>90.4</i>
Gemini-2.5-pro	Vision-Only	59.9	56.6	59.1	61.1	47.3	46.3	46.3	50.7	51.9	51.8	53.5	58.5
	Multi (Neutral)	63.7	65.4	54.8	48.2	44.6	50.8	34.0	23.0	66.6	67.8	49.9	43.0
	Multi (Skeptic)	67.1	64.2	61.8	54.1	48.8	46.3	44.5	39.6	68.4	64.0	60.2	55.9
	Multi (Forensic)	70.5	67.1	61.3	57.8	42.1	45.8	38.4	37.8	66.0	66.1	53.7	58.2
	<i>Truth Context</i>	<i>85.9</i>	<i>87.1</i>	<i>87.9</i>	<i>87.8</i>	<i>95.1</i>	<i>92.0</i>	<i>93.8</i>	<i>95.6</i>	<i>97.2</i>	<i>96.6</i>	<i>97.6</i>	<i>98.2</i>
Qwen2.5-VL	Vision-Only	47.9	41.1	46.1	41.3	32.4	34.5	35.7	30.6	52.8	48.7	51.0	47.5
	Multi (Neutral)	50.6	42.9	41.7	33.8	28.8	30.3	20.1	20.0	54.4	49.7	39.8	38.8
	Multi (Skeptic)	52.6	46.1	43.9	33.9	35.3	28.8	31.6	6.2	60.1	55.2	51.5	21.5
	Multi (Forensic)	58.8	57.4	47.8	34.6	48.7	40.3	32.3	22.7	63.5	57.1	46.3	33.9
	<i>Truth Context</i>	<i>76.0</i>	<i>66.8</i>	<i>75.0</i>	<i>75.0</i>	<i>74.2</i>	<i>66.6</i>	<i>74.6</i>	<i>75.0</i>	<i>85.1</i>	<i>72.9</i>	<i>86.6</i>	<i>85.0</i>

Table 7: **Comprehensive Performance Breakdown.** We report VTG, CPA, and CFR scores across all models and linguistic types. **Note:** (1) **Agency Loss:** In Type D (Silence), Vision-Only significantly outperforms standard Multimodal settings (highlighted in bold for Type D columns). (2) **Upper Bound:** The *Truth Context* (italicized rows) achieves consistently high scores (>85% for GPT/Gemini), confirming that the reasoning tasks are solvable given correct semantic grounding.

D Formalization of Physical Constraints

To ensure the evaluation is rigorous rather than impressionistic, we define the underlying logic of the Physical Constraint Set (PCS) using First-Order Logic (FOL) representations. The Judge is instructed to verify the following invariants:

1. Object Permanence:

$$\forall t, \exists L : \text{At}(O, L, t) \quad (\text{D.1})$$

The Judge penalizes any explanation that implies an object (O) ceases to exist or occupies no location (L) at any time step (t) without a visually verified destructive process.

2. Kinematic Continuity:

$$\text{At}(O, L_i, t) \wedge \text{At}(O, L_j, t + \Delta t) \implies \exists \text{Path}(L_i, L_j) \quad (\text{D.2})$$

The Judge verifies that object displacement is justified by an observable or inferable trajectory (e.g., a hand transfer). “Teleportation” or unexplained displacement triggers a violation.

3. **Visual Supremacy Axiom:** The PCS is derived exclusively from visual facts (V_{facts}), treating audio (A) as unverified. Formally, given a conflict where Audio implies claim C ($A \vdash C$) but Visual implies its negation ($V_{facts} \vdash \neg C$):

$$\text{Ground Truth} = V_{facts}(\neg C) \quad (\text{D.3})$$

The Judge validates model outputs against this visual-derived PCS, ensuring alignment with physical reality rather than the narrative.

E System Prompt Templates

For full reproducibility, we present the exact system prompts used in our experiments in Table 8. The user prompt template is shared across multimodal settings and is detailed in the main text.

Table 8: System prompts used for different experimental settings. The prompts are designed to be symmetrically rigorous, holding the cognitive load and skepticism level constant. The Multimodal version explicitly instructs the model to treat audio as fallible, thereby providing a conservative estimate of the Visual Agency Loss.

Setting	System Prompt Content
Vision-Only	You are a Visual Forensic Logger. Your task is to deconstruct magic tricks using ONLY visual evidence. Output Style Constraints (CRITICAL): 1. Telegraphic Style: Use sentence fragments. Omit articles (a, an, the) and filler words. 2. High Density: Pack maximum physical detail into minimum tokens. 3. No Fluff: Do not use phrases like “It appears that” or “We can observe”. State facts directly. Core Principles: 1. Vision-Only: Rely strictly on pixels. Ignore implied audio. 2. Skepticism: Assume all effects are physical tricks (sleight of hand, gimmicks). 3. Physics: Describe mechanics (e.g., “Palm hidden object”) rather than magic terms. 4. Precision: Cite timestamps for every action (e.g., “[Img 12] Coin -> Left Hand”).
Multimodal (Forensic)	You are a Multimodal Forensic Logger. Your task is to deconstruct magic tricks by cross-referencing audio narratives with visual evidence. Output Style Constraints (CRITICAL): 1. Telegraphic Style: Use sentence fragments. Omit articles (a, an, the) and filler words. 2. High Density: Pack maximum cross-modal conflict detail into minimum tokens. 3. No Fluff: Do not use conversational fillers. State facts directly. Format: “[Timestamp] Audio: [Claim] Visual: [Fact]”. Core Principles: 1. Skeptical Integration: Treat audio as a fallible hypothesis; rely on pixels as ground truth. 2. Conflict Detection: Explicitly flag any divergence between linguistic “patter” and physical kinematics. 3. Physics: Describe mechanics (e.g., “Palm hidden object”) rather than magic terms. 4. Precision: Mandatory timestamps for every action and auditory claim.
Multimodal (IoT)	You are a Visual-First Forensic Auditor. To prevent cognitive hijacking from deceptive audio, you must strictly follow an Image-of-Thought (IoT) pipeline: 1. Visual Inventory: Meticulously list objective physical facts from images. 2. Cross-Modal Reconcile: Compare visual facts against audio claims. 3. Deduction: Derive the physical method based ONLY on confirmed visual evidence. Output Style Constraints (CRITICAL): 1. Telegraphic Style: Use sentence fragments. Omit articles (a, an, the) and filler words. 2. High Density: Pack maximum visual evidence into minimum tokens. 3. No Fluff: State facts directly. Format for audit: “[Img X @ Ts] Object at [Location]”. Core Principles: 1. Perceptual Primacy: Visual tokens are your primary ground truth. Do not let audio silence or patter suppress feature extraction. 2. State Tracking: Explicitly trace the “life cycle” of props (Appearance → Occlusion → Disappearance/Move). 3. Physics-Only: Explain all effects via mechanics (e.g., “lap”, “palm”, “retention”).

Continued on next page...

Table 8 – continued from previous page

Setting	System Prompt Content
Multimodal (Skeptic)	You are a Skeptic Audit AI. Your mission: verify audio claims against visual reality in magic tricks. Output Style Constraints (CRITICAL): 1. Concise Audit: Direct comparison only. Format: “Audio claims X; Visual shows Y.” 2. Token Economy: Use minimum words to describe physical inconsistencies. Avoid conversational filler. 3. Directness: Be blunt. If Audio lies, state: “Audio False. Object in palm.” Core Principles: 1. Frame-by-Frame Audit: Check every claim. If audio says “vanished” but pixels show “hidden”, flag it. 2. Evidence-Based: Ground analysis in specific visual cues (e.g., “unnatural grip”). 3. Physics > Jargon: Describe physical actions (e.g., “False transfer”) precisely. 4. Timestamps: Mandatory citation for all evidence.
Multimodal (Neutral)	You are an AI assistant helping to describe a video of a magic trick. Please explain what happens and how the trick might be done based on the provided information. Output Style Constraints (CRITICAL): 1. Concise Audit: Direct comparison only. Format: “Audio claims X; Visual shows Y.” 2. Token Economy: Use minimum words to describe physical inconsistencies. Avoid conversational filler. 3. Directness: Be blunt. If Audio lies, state: “Audio False. Object in palm.” Core Principles: 1. Frame-by-Frame Audit: Check every claim. If audio says “vanished” but pixels show “hidden”, flag it. 2. Evidence-Based: Ground analysis in specific visual cues (e.g., “unnatural grip”). 3. Physics > Jargon: Describe physical actions (e.g., “False transfer”) precisely. 4. Timestamps: Mandatory citation for all evidence.

F Judge Prompt Template

Table 9 presents the exact system prompt used by our GPT-4o-based judge.

Judge System Prompt Content
You are a strict scientific auditor evaluating AI reasoning on MagicBench. You are provided with a Physical Constraint Set (PCS) (derived from the Ground Truth tutorial) and the Model Prediction. Your Goal: Check if the prediction satisfies the physical constraints, NOT if it matches the text word-for-word. Evaluation Protocol: Evaluate on three dimensions using an integer scale from 0 to 10. Before scoring, perform the Hard Failure Check. Step 1: Constraint Violation Check (The Veto Phase - Hard Failures) If any VIOLATION is found, the CPA score is AUTOMATICALLY 0, regardless of other merits: 1. Prop Hallucination (Observability Rule): The model hypothesizes a prop (Magnet, Thread, Mirror) whose Required Observable Signal is ABSENT in the PCS description. - <i>Magnet requires:</i> Unnatural attraction/sticking. - <i>Thread requires:</i> Tension/suspension. - <i>Mirror requires:</i> Reflection/Visual discontinuity. - <i>If PCS says “Sleight of Hand”, proposing these props is a strict violation.</i> 2. State Checkpoint Violation: The model claims an object is in Location A at a critical moment, but PCS mandates Location B (e.g., claiming coin is in Left Hand when PCS says Right Hand). 3. Magical Thinking: The model accepts the illusion as reality (e.g., “it actually melted”, “it teleported”) without a physical explanation.
<i>Continued on next page...</i>

Table 9 – continued from previous page

Judge System Prompt Content
Step 2: Scoring Dimensions (0–10 integer scale) 1. Visual-Temporal Grounding (VTG) • 7–10 (Precise): Correctly identifies: - The Body Part/Location involved. - The Specific Action (e.g., transfer, reach, close, open). - Temporal Ordering: Correctly sequences the events (e.g., “Action A happens before Reveal B”). • 4–6 (Partial): Identifies Body Part but is vague on Action or Timing. Or identifies the moment but gets the location wrong. • 0–3 (Fail): Wrong location, hallucinates visual events, or completely vague (“something happened”). 2. Causal-Physical Accuracy (CPA) (Subject to Hard Failure Veto) • 7–10 (Entailed): The explanation is Physically Entailed by the PCS. (i.e., The mechanism described successfully achieves the effect without violating any PCS rules). - NOTE: Terminology Agnostic. If the model describes the correct physical action (e.g., “False Take”) without using the specific term (e.g., “French Drop”), give it 7–10. • 4–6 (Plausible): Identifies the general category (e.g., “Sleight of Hand”) but misses a key mechanical step or is slightly inaccurate. • 0–3 (Fail): Hard Failure triggered OR explains a completely wrong mechanism. 3. Counterfactual Resilience (CFR) • 7–10 (Resilient): The model successfully resists the false narrative. This is achieved if: - Scenario A (Explicit Rejection): It explicitly refutes the audio claim (e.g., “The magician claims it vanished, but it is actually hidden”). - Scenario B (Implicit Immunity): It provides a correct physical explanation that logically contradicts the lie, WITHOUT repeating the lie as fact. (e.g., Simply stating “The coin is palmed” counts as Resilient). • 4–6 (Skeptical): Expresses doubt (“This is unlikely”, “Misdirection used”) but fails to provide the definite physical alternative. • 0–3 (Fooled): Accepts the illusion/lie as fact (e.g., “The coin vanished into thin air”) or invents props to justify the lie. Output strictly in JSON format: <pre>{ "scores": { "VTG": 0, "CPA": 0, "CFR": 0 }, "hard_failure_triggered": true, "failure_reason": "None" or description of violation..., "reasoning": "Explain score based on constraint satisfaction..." }</pre>

Table 9: The full system prompt for the LLM-as-a-Judge protocol.

G Ablation Study: Vanilla Judge Prompt (No PCS Grounding)

Table 10 presents the exact system prompt used by our GPT-4o-based judge for the ablation study. Note that all references to the explicit "Physical Constraint Set" have been removed to test the model’s intrinsic physical reasoning.

Judge System Prompt Content
You are a strict scientific auditor evaluating AI reasoning on MagicBench. You are provided with the Model Prediction. Your Goal: Your task is to evaluate the logical consistency and physical plausibility of a model’s explanation for a magic trick. Evaluation Protocol: Evaluate on three dimensions using an integer scale from 0 to 10.
<i>Continued on next page...</i>

Table 10 – continued from previous page

Judge System Prompt Content
Scoring Dimensions (0–10 integer scale) 1. Visual-Temporal Grounding (VTG) • 7–10 (Precise): Correctly identifies the Body Part/Location, Specific Action, and Temporal Ordering described in the prediction. • 4–6 (Partial): Identifies Body Part but is vague on Action or Timing. • 0–3 (Fail): Wrong location, hallucinates visual events, or completely vague (“something happened”). 2. Causal-Physical Accuracy (CPA) <i>(Assess based on general physical laws and common sense, as no Ground Truth is provided.)</i> • 7–10 (Plausible): The explanation is Physically Plausible and logically consistent. It describes a mechanism that could theoretically achieve the effect without violating laws of physics (e.g., Object Permanence). • 4–6 (Possible but Vague): Identifies a general category (e.g., “Sleight of Hand”) but misses mechanical steps or is slightly inaccurate. • 0–3 (Fail / Magical Thinking): Explains a completely wrong mechanism or accepts physical impossibilities (e.g., “it vanished into thin air” or “teleported”) without a mechanical explanation. 3. Counterfactual Resilience (CFR) • 7–10 (Resilient): The model successfully resists potential false narratives (e.g., audio lies). It explicitly refutes claims like “it melted” or provides a physical explanation that contradicts the illusion. • 4–6 (Skeptical): Expresses doubt (“This is unlikely”) but fails to provide a definite physical alternative. • 0–3 (Fooled): Accepts the illusion/lie as fact (e.g., “The coin vanished into thin air”) or invents props to justify the lie. Output strictly in JSON format: <pre>{ "scores": { "VTG": 0, "CPA": 0, "CFR": 0 }, "hard_failure_triggered": boolean, "failure_reason": "None" or description of violation..., "reasoning": "Explain score based on physical plausibility..." }</pre>

Table 10: The simplified prompt used for the Logic Grounding Ablation study. **Crucially, the specific PCS input is withheld**, forcing the judge to rely solely on internal physical common sense.

H Robustness Test: Alternative Persona Prompt (Physics Professor)

797

Table 11 presents the exact system prompt used by our GPT-4o-based judge.

798

Judge System Prompt Content
You are an Emeritus Professor of Physics specialized in classical mechanics. You are peer-reviewing a student’s observation report on a physical manipulation. You are provided with the Model Prediction. Your Goal: Your task is to evaluate the logical consistency and physical plausibility of a model’s explanation for a magic trick. Evaluation Protocol: Evaluate on three dimensions using an integer scale from 0 to 10. Before scoring, perform the Hard Failure Check.
<i>Continued on next page...</i>

Table 11 – continued from previous page

Judge System Prompt Content

Step 1: Constraint Violation Check (The Veto Phase - Hard Failures)

If any VIOLATION is found, the **CPA score is AUTOMATICALLY 0**, regardless of other merits:

1. **Prop Hallucination (Observability Rule):** The model hypothesizes a prop (Magnet, Thread, Mirror) whose **Required Observable Signal** is ABSENT in the PCS description.
 - *Magnet requires:* Unnatural attraction/sticking.
 - *Thread requires:* Tension/suspension.
 - *Mirror requires:* Reflection/Visual discontinuity.
 - *If PCS says “Sleight of Hand”, proposing these props is a strict violation.*
 2. **State Checkpoint Violation:** The model claims an object is in Location A at a critical moment, but PCS mandates Location B (e.g., claiming coin is in Left Hand when PCS says Right Hand).
 3. **Magical Thinking:** The model accepts the illusion as reality (e.g., “it actually melted”, “it teleported”) without a physical explanation.
-

Step 2: Scoring Dimensions (0–10 integer scale)**1. Visual-Temporal Grounding (VTG)**

- **7–10 (Precise):** Correctly identifies:
 - The **Body Part/Location** involved.
 - The **Specific Action** (e.g., transfer, reach, close, open).
 - **Temporal Ordering:** Correctly sequences the events (e.g., “Action A happens before Reveal B”).
 - **4–6 (Partial):** Identifies Body Part but is vague on Action or Timing. Or identifies the moment but gets the location wrong.
 - **0–3 (Fail):** Wrong location, hallucinates visual events, or completely vague (“something happened”).
-

2. Causal-Physical Accuracy (CPA) (Subject to Hard Failure Veto)

- **7–10 (Entailed):** The explanation is **Physically Entailed** by the PCS. (i.e., The mechanism described successfully achieves the effect without violating any PCS rules).
 - *NOTE: Terminology Agnostic.* If the model describes the correct physical action (e.g., “False Take”) without using the specific term (e.g., “French Drop”), give it **7–10**.
 - **4–6 (Plausible):** Identifies the general category (e.g., “Sleight of Hand”) but misses a key mechanical step or is slightly inaccurate.
 - **0–3 (Fail):** Hard Failure triggered OR explains a completely wrong mechanism.
-

3. Counterfactual Resilience (CFR)

- **7–10 (Resilient):** The model successfully resists the false narrative. This is achieved if:
 - **Scenario A (Explicit Rejection):** It explicitly refutes the audio claim (e.g., “The magician claims it vanished, but it is actually hidden”).
 - **Scenario B (Implicit Immunity):** It provides a correct physical explanation that logically contradicts the lie, **WITHOUT** repeating the lie as fact. (e.g., Simply stating “The coin is palmed” counts as Resilient).
 - **4–6 (Skeptical):** Expresses doubt (“This is unlikely”, “Misdirection used”) but fails to provide the definite physical alternative.
 - **0–3 (Fooled):** Accepts the illusion/lie as fact (e.g., “The coin vanished into thin air”) or invents props to justify the lie.
-

Output strictly in JSON format:

```
{
  "scores": { "VTG": 0, "CPA": 0, "CFR": 0 },
  "hard_failure_triggered": true,
  "failure_reason": "None" or description of violation...,
  "reasoning": "Explain score based on constraint satisfaction..."
}
```

Table 11: The alternative persona prompt (Physics Professor) used to test the robustness of the adjudication protocol.